Proximal Policy Optimization with Elastic Weight Consolidation for Adaptive Ocean Thermal Energy Conversion Control

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Abstract*—Ocean Thermal Energy Conversion systems deployed in tropical marine environments face strongly non-stationary operating conditions driven by spatiotemporal variations in sea surface temperature. These variations induce shifting system dynamics that cause conventional and standard reinforcement learning controllers to suffer catastrophic forgetting—a progressive loss of previously learned control knowledge when adapting sequentially to new thermal regimes. This work proposes a continuous reinforcement learning framework for adaptive supervisory control of open-cycle Ocean Thermal Energy Conversion systems. Proximal Policy Optimization is augmented with Elastic Weight Consolidation regularization to constrain parameter updates according to Fisher Information-weighted importance scores. Real-world sea surface temperature data from NetCDF ocean datasets is preprocessed into four sequential thermal regimes representing distinct ocean operating conditions. The control problem is formulated as a continuous-action Markov decision process in which the agent regulates the flow rate, pressure, and turbine intensity to maximize net power output. Sequential training across four thermal regimes showed that the proposed framework achieved average rewards of 0.2909, 0.0219, and 0.1530 in the regimes T1–T3, compared to -0.2454, -0.0899, and -0.0805 for the baseline, representing an improvement 18.6% in stability of T1 and recovery of positive rewards in T2 and T3 where the baseline produced negative returns. The baseline exhibited clear catastrophic forgetting on earlier tasks; the proposed method reduced this degradation through selective weight regularization. A trade-off was observed in the final regime, where unconstrained learning retained greater plasticity. These results demonstrate that Elastic Weight Consolidation-augmented continual reinforcement learning provides a viable pathway toward knowledge-retentive, adaptive control of renewable marine energy systems under real-world non-stationary ocean conditions.

***Index Terms*—ocean thermal energy conversion; continual learning; proximal policy optimization; elastic weight consolidation; catastrophic forgetting; non-stationary environments; stability-plasticity trade-off; marine energy systems**

I. Introduction

Lately, everyone seems to be talking about clean energy, and ocean-based renewables are finally getting their moment. Ocean Thermal Energy Conversion, or OTEC, really stands out. It uses the natural temperature difference between the sun-warmed surface and the cold, deep water in the tropics to keep power flowing 24/7. That alone puts it ahead of solar and wind, which can stop and start whenever the weather changes. Open-cycle OTEC, where the seawater itself does the work, fits perfectly in tropical and subtropical regions. It's a simpler design, and as a bonus, you get fresh water from the process. With energy demand rising, OTEC has real potential—especially for coastal communities—and honestly, it's been hiding in plain sight. Still, running an OTEC system isn't easy. The ocean's always moving. You get different currents, big seasonal swings, and changes year to year, so the vital temperature gradient keeps shifting. That means the system's efficiency goes up and down. Most control solutions—basic PID controllers and rule-based systems—do fine when things are steady, but as soon as conditions shift, engineers end up retuning everything by hand. That's slow, expensive, and doesn't scale well if you want these systems to run mostly on their own in the real world. Reinforcement learning, or RL, looks like a way out. These algorithms don't need perfect mathematical models. They pick smart strategies just by interacting with the system, learning as they go. One of the big names here is Proximal Policy Optimization (PPO). It's already shown it can solve tricky energy problems, especially when things don't fall into neat categories. But there's a catch: regular RL agents forget what they learned. If you retrain them on new ocean conditions, they tend to lose the stuff they picked up before—a problem known as "catastrophic forgetting." Continual reinforcement learning tries to fix this. It teaches agents to hold on to what matters from past tasks while picking up new skills. One standout technique for this is Elastic Weight Consolidation (EWC). EWC helps the agent remember which bits of its experience were important and makes sure those parts don't get overwritten when something new comes along. Surprisingly, nobody's really tried bringing continual reinforcement learning into OTEC—especially not for handling all the twists and turns in sea surface temperatures. Most research sticks to a single, fixed environment, or just glosses over how much RL agents forget when things change. That's a problem if you're serious about running these systems in an unpredictable ocean. You can't afford to lose old skills every time the water changes. That's the gap this paper tries to close. It's the first to use continual reinforcement learning for OTEC supervisory control, focusing on the messy, real-life temperature shifts that open-cycle systems face. We pull actual sea surface temperature data from NetCDF ocean datasets and split it into four realistic, back-to-back scenarios.

Then we train a PPO controller, powered up with Elastic Weight Consolidation, and run it through the four regimes one after another. We watch learning curves, track how much the agent forgets as conditions shift, and look at how it balances staying flexible with holding onto hard-earned skills. The goal is to make OTEC controllers smarter and more reliable—ready to handle whatever the ocean throws their way. The primary contributions of this work are:

- **A continual reinforcement learning control framework** — the first application of Proximal Policy Optimization with Elastic Weight Consolidation to open-cycle Ocean Thermal Energy Conversion supervisory control, formulated as a continuous-action Markov decision process across four non-stationary thermal regimes.
- **A data-driven Ocean Thermal Energy Conversion simulation environment** — a custom Gym-compatible environment built on real-world NetCDF sea surface temperature data, with regime-specific reward functions that encode distinct operational constraints.
- **A comprehensive continual learning evaluation** — quantitative analysis of catastrophic forgetting, knowledge retention, and the stability–plasticity trade-off across sequential tasks, benchmarked against a standard Proximal Policy Optimization baseline.

The remainder of this paper is organized as follows. Section II reviews related work on Ocean Thermal Energy Conversion control, reinforcement learning, and continual learning. Section III describes the methodology, including the environment design, Markov decision process formulation, and Elastic Weight Consolidation integration. Section IV presents the experimental setup and evaluation protocol. Section V reports results and Section VI discusses findings, limitations, and implications for real-world deployment. Section VII concludes the paper.

II. Related Work

Research relevant to this work spans three interconnected domains: Ocean Thermal Energy Conversion control, reinforcement learning for energy systems, and continual learning for non-stationary environments. This section synthesizes existing contributions across these areas, identifies their limitations, and positions the present work within the broader literature.

A. Ocean Thermal Energy Conversion Systems and Control

Ocean Thermal Energy Conversion has been studied as a viable renewable energy source since the 1970s, with foundational thermodynamic analyses establishing the theoretical efficiency bounds of both open-cycle and closed-cycle configurations [1]. Open-cycle systems, in which warm seawater flashes into steam through a low-pressure turbine, offer the additional benefit of simultaneous freshwater desalination, making them particularly attractive for tropical island deployment [4]. Control of Ocean Thermal Energy Conversion systems has historically relied on proportional-integral-derivative regulators and rule-based supervisory schemes. Lennard [26] demonstrated that fixed-gain controllers are sufficient under stable tropical conditions but fail under inter-seasonal sea surface temperature drift exceeding 2°C. More recently, Adiputra et al. [2] performed a preliminary design study for a 100 kW open-cycle plant and identified adaptive flow-rate control as a critical unsolved challenge under dynamic ocean conditions. Xiang et al. [27] proposed a model-predictive control scheme for closed-cycle Ocean Thermal Energy Conversion that improved power output stability by 12% under simulated seasonal variation, but their approach assumed full knowledge of the thermal gradient model and did not address sequential regime shifts. Collectively, these studies confirm that existing Ocean Thermal Energy Conversion controllers are designed for quasi-static conditions and lack mechanisms for autonomous adaptation under non-stationary sea surface temperature dynamics.

B. Reinforcement Learning for Energy System Control

Reinforcement learning has gained significant traction as a model-free control paradigm for complex energy systems. François- Lavet et al. [28] provided an early demonstration of deep reinforcement learning for demand-response optimization in smart grids, establishing that policy-gradient methods can handle the high-dimensional, continuous action spaces characteristic of energy systems. Subsequent work by Zhang et al. [10] applied Proximal Policy Optimization to microgrid energy management, achieving a 17% reduction in operating cost compared to model-predictive control baselines, with the actor–critic architecture providing stable convergence under noisy sensor inputs.

In the marine energy domain, Anderlini et al. [9] applied deep reinforcement learning to wave energy converter control, demonstrating that learned policies outperform classical reactive controllers under irregular wave conditions. However, all of the above studies train agents in stationary environments — a single fixed operating regime — and do not evaluate performance degradation when conditions shift sequentially. This is a critical gap for Ocean Thermal Energy Conversion, where sea surface temperature regimes change across seasons and climate cycles.

C. Continual Learning and Catastrophic Forgetting

Continual learning, also termed lifelong learning, addresses the problem of sequential task acquisition without forgetting prior knowledge. The dominant failure mode is catastrophic forgetting, in which gradient-based updates for a new task overwrite weights critical to earlier tasks. Three families of mitigation strategies have emerged in the literature: regularization-based, replay-based, and architecture-based methods.

Kirkpatrick et al. [12] introduced Elastic Weight Consolidation, which estimates parameter importance via the Fisher Information Matrix and adds a quadratic penalty term to the loss function, preventing large updates to task-critical weights. Evaluated on supervised sequential classification tasks, Elastic Weight Consolidation reduced forgetting by over 40% compared to unconstrained fine-tuning. Zenke et al. [13] proposed

Synaptic Intelligence, an online variant that accumulates importance scores during training rather than post-hoc, offering computational advantages for long task sequences. Replay-based methods such as Gradient Episodic Memory [14] store a small buffer of past experiences and enforce non-interference constraints during new task learning, achieving strong retention at the cost of memory overhead. Architecture-based approaches such as Progressive Neural Networks allocate separate capacity for each task, eliminating forgetting entirely but scaling poorly with the number of tasks.

Among these, Elastic Weight Consolidation remains the most widely adopted for control applications due to its low memory footprint and compatibility with any gradient-based optimizer, making it the natural choice for integration with Proximal Policy Optimization in the present work.

D. Continual Reinforcement Learning for Non-Stationary Control

The integration of continual learning with reinforcement learning for non-stationary control has gained momentum in recent years. Rolnick et al. [18] demonstrated that experience replay combined with policy-gradient methods can sustain performance across shifting Atari game distributions, but their approach requires storing full trajectory buffers, which is impractical for embedded marine energy controllers. Khetarpal et al. [19] provided a theoretical framework for continual reinforcement learning in non-stationary Markov decision processes, identifying the stability–plasticity trade-off as the central design challenge and calling for domain-specific benchmarks beyond robotics and games.

In the power systems domain, Cao et al. [20] applied continual reinforcement learning to building energy management under shifting occupancy patterns, showing that Elastic Weight Consolidation-augmented agents retained 78% of prior-task performance compared to 51% for unconstrained fine-tuning. However, their environment involved discrete action spaces and single-building scenarios, which do not capture the continuous multi-variable control demands of Ocean Thermal Energy Conversion supervisory systems.

E. Gap Analysis and Positioning

Table I summarizes the key characteristics of representative works across these domains. Several limitations emerge consistently. First, no prior study applies continual reinforcement learning to Ocean Thermal Energy Conversion control in any configuration. Second, existing Ocean Thermal Energy Conversion control works do not evaluate sequential regime shifts or knowledge retention. Third, continual reinforcement learning studies in energy systems are limited to discrete-action, single- building scenarios and do not address marine or thermal-gradient environments. Fourth, stability–plasticity trade-off analysis is absent from all Ocean Thermal Energy Conversion and marine energy control studies reviewed.

The present work directly addresses all four gaps by proposing the first continual reinforcement learning framework for open-cycle Ocean Thermal Energy Conversion supervisory control, formulated as a continuous-action Markov decision process trained sequentially across four real-world sea surface temperature regimes, with explicit evaluation of catastrophic forgetting and the stability– plasticity trade-off.

III. Methodology

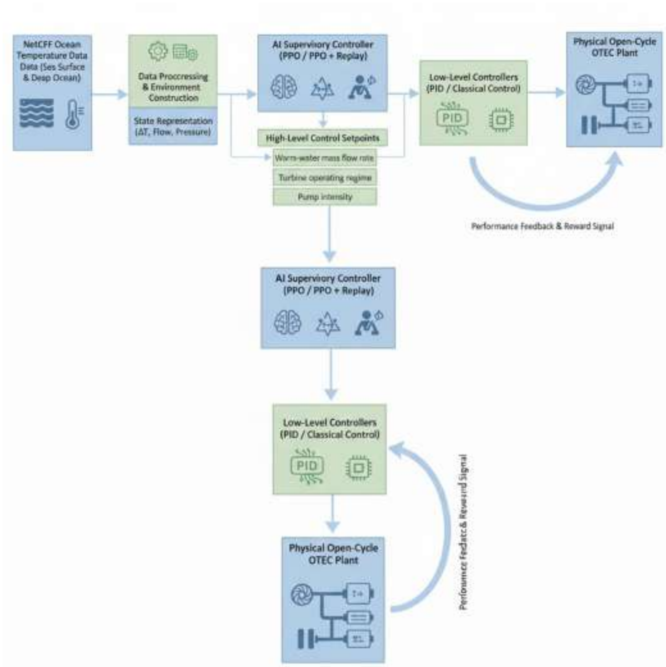

Fig. 1. Overall reinforcement learning-based supervisory control pipeline for the open-cycle OTEC system.

This section presents the complete formulation of the continual reinforcement learning framework for open-cycle Ocean Thermal Energy Conversion supervisory control. The control problem is first cast as a Markov decision process, followed by descriptions of the environment design, state and action spaces, reward function, sequential learning protocol, and the Proximal Policy Optimization with Elastic Weight Consolidation training algorithm.

A. Problem Formulation as a Markov Decision Process

The supervisory control problem is formulated as a discrete-time, continuous-action Markov decision process defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma)$, where:

- $\mathcal{S} \subseteq \mathbb{R}^4$ is the continuous state space of sea surface temperature statistics,
- $\mathcal{A} \subseteq [-1, 1]^3$ is the continuous action space of normalized control variables,
- $\mathcal{T} : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ is the environment transition function,
- $\mathcal{R} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function encoding net power generation and control effort,
- $\gamma = 0.99$ is the discount factor governing the agent's temporal horizon.

The agent observes state $\mathbf{s}_t \in \mathcal{S}$ at each timestep t, selects action $\mathbf{a}_t \in \mathcal{A}$ according to policy $\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)$, receives scalar

TABLE I
COMPARISON OF REPRESENTATIVE RELATED WORKS

Reference	**Domain**	**RL Method**	**Continual Learning**	**Non-stationary Env.**	**OTEC Control**	**Forgetting Analysis**
Adiputra et al. [2]	OTEC design	None	×	×	✓	×
Xiang et al. [27]	OTEC control	MPC	×	Partial	✓	×
Zhang et al. [10]	Microgrid	PPO	×	×	×	×
Anderlini et al. [9]	Wave energy	DRL	×	×	×	×
Kirkpatrick et al. [12]	Vision/NLP	None	EWC	Seq. tasks	×	✓
Zenke et al. [13]	Vision	None	SI	Seq. tasks	×	✓
Rolnick et al. [31]	Games	DQN	Replay	✓	×	Partial
Cao et al. [20]	Building energy	DQN	EWC	✓	×	✓
Khetarpal et al. [19]	Theory	General	General	✓	×	✓
This work	**OTEC control**	**PPO+EWC**	**EWC**	✓	✓	✓

reward r_t, and transitions to $\mathbf{s}_{t+1}$. The objective is to find parameters θ^* that maximize the expected discounted return:

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t r_t\right] \tag{1}$$

B. Environment Design

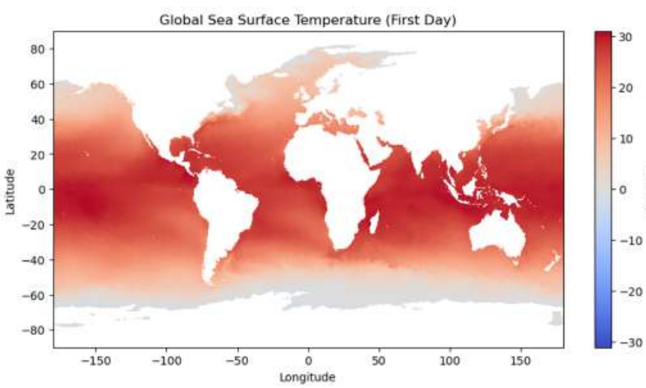

Fig. 2. Global SST distribution used for constructing environmental regimes.

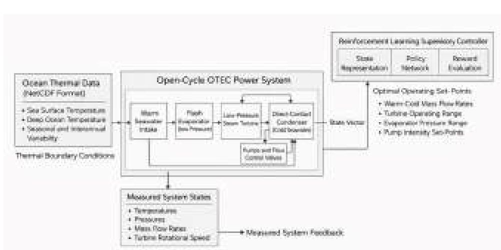

Fig. 3. Physical open-cycle OTEC system model.

The open-cycle Ocean Thermal Energy Conversion system is implemented as a custom `gym.Env`-compatible environment following the standard `reset()/step()` interface [**?**]. Each episode consists of a fixed-length sequence of sea surface temperature observations drawn from the active thermal regime T_k. The environment computes the instantaneous reward using the simplified power model described in Section III-E, without requiring high-fidelity thermodynamic simulation, thereby maintaining computational tractability for large-scale reinforcement learning training.

C. State Representation

The raw sea surface temperature data from NetCDF ocean datasets is compressed into a four-dimensional statistical feature vector. At each timestep t, the state is defined as:

$$\mathbf{s}_t = \left[\mu_{\text{SST}},\ \sigma_{\text{SST}},\ \min(\text{SST}),\ \max(\text{SST})\right]^\top \tag{2}$$

where μ_{SST} is the mean sea surface temperature, σ_{SST} is the standard deviation capturing thermal variability, and $\min(\text{SST})$, $\max(\text{SST})$ define the boundary conditions of the observed distribution. All features are normalized to zero mean and unit variance computed over the training split of each regime, ensuring numerical stability and compatibility with neural network policies.

D. Action Space

The supervisory control action is a three-dimensional continuous vector:

$$\mathbf{a}_t = \left[f_t,\ p_t,\ \tau_t\right]^\top \in [-1, 1]^3 \tag{3}$$

where f_t is the normalized seawater flow rate, p_t is the normalized system pressure, and τ_t is the turbine intensity. All three components are dimensionless supervisory signals scaled to physical operating ranges during environment interaction. Using a normalized action space stabilizes policy gradient updates and decouples learning dynamics from physical unit scaling.

E. Reward Function

The instantaneous power output is modelled as:

$$P_t = \Delta T_t \cdot f_t \cdot \tau_t - c \cdot f_t^2 \tag{4}$$

where ΔT_t is the thermal gradient approximated from the sea surface temperature statistics at timestep t, and $c > 0$ is a pumping-loss coefficient. The base reward penalizes excessive flow and pressure:

$$r_t = P_t - \alpha f_t - \beta p_t \tag{5}$$

with $\alpha, \beta > 0$ encouraging smoother control trajectories. To reflect distinct operational economics across thermal regimes, regime-specific reward functions are defined as:

$$r_t^{(k)} = \begin{cases} P_t & k = T_1 \\ P_t - 0.5\, f_t & k = T_2 \\ P_t - 0.5\, p_t & k = T_3 \\ P_t - 0.7\,(f_t + p_t) & k = T_4 \end{cases} \tag{6}$$

Each regime $k \in \{T_1, T_2, T_3, T_4\}$ progressively increases the penalty on control effort, creating a non-stationary reward landscape that demands policy adaptation across the task sequence.

F. Sequential Learning Protocol

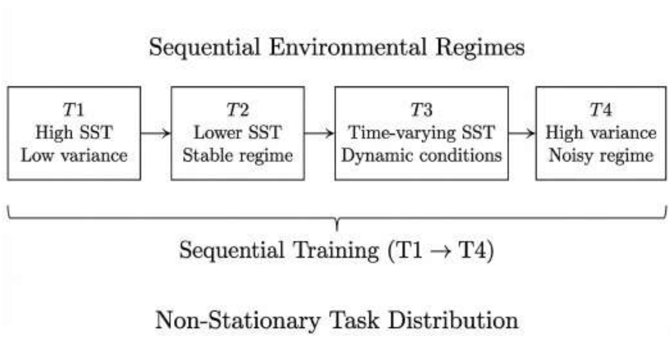

Fig. 4. Sequential SST regime training timeline ($T_1 \to T_4$).

Non-stationary ocean conditions are modelled as a task sequence:

$$\mathcal{T} = \{T_1, T_2, T_3, T_4\} \tag{7}$$

Training proceeds sequentially, with model parameters carried forward between tasks:

$$\theta_1 \;\rightarrow\; \theta_2 \;\rightarrow\; \theta_3 \;\rightarrow\; \theta_4 \tag{8}$$

where θ_k denotes the policy parameters after completing training on task T_k. Without a forgetting-mitigation mechanism, gradient updates for T_{k+1} overwrite weights critical to T_k, causing catastrophic forgetting.

G. Proximal Policy Optimization Baseline

Proximal Policy Optimization employs an actor–critic architecture in which the actor π_θ outputs a Gaussian action distribution and the critic V_ϕ estimates the state-value function. The clipped surrogate objective is:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\; \text{clip}(r_t(\theta),\, 1{-}\epsilon,\, 1{+}\epsilon)\,\hat{A}_t\Big)\Big] \tag{9}$$

where the probability ratio is:

$$r_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{s}_t)} \tag{10}$$

and $\hat{A}_t$ is the Generalized Advantage Estimate with $\lambda_{\text{GAE}} = 0.95$. The clip parameter $\epsilon = 0.2$ limits destructive policy updates, providing stable convergence in continuous action spaces.

H. Elastic Weight Consolidation Integration

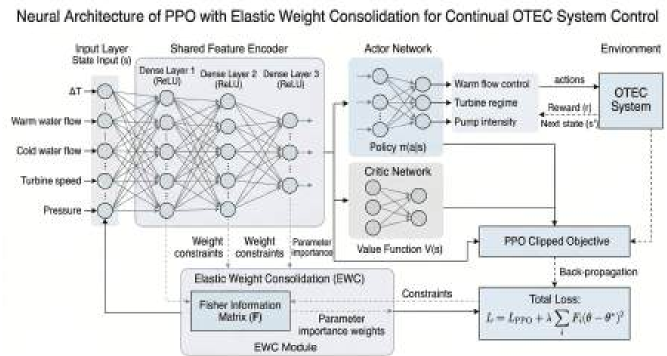

Fig. 5. Proposed PPO+EWC architecture with Fisher-based regularization.

Elastic Weight Consolidation mitigates catastrophic forgetting by adding a quadratic regularization term to the Proximal Policy Optimization objective. Parameter importance for task T_k is estimated via the diagonal of the Fisher Information Matrix:

$$F_i^{(k)} = \mathbb{E}\left[\left(\frac{\partial \log \pi_\theta(\mathbf{a} \mid \mathbf{s})}{\partial \theta_i}\right)^2\right] \tag{11}$$

The combined loss for training on task T_{k+1} is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{PPO}} + \lambda \sum_i F_i^{(k)} \big(\theta_i - \theta_i^*\big)^2 \tag{12}$$

where θ_i^* are the parameters consolidated after task T_k and $\lambda = 5000$ controls regularization strength. Large $F_i^{(k)}$ values indicate parameters critical to prior tasks; the penalty term prevents these from drifting substantially during adaptation to new regimes.

I. Stability–Plasticity Trade-off Metric

The tension between knowledge retention and adaptability is quantified using two scalar performance metrics:

$$S = R_{T_1}, \qquad P = R_{T_4} \tag{13}$$

where S measures stability as performance on the earliest task after all training is complete, and P measures plasticity as performance on the most recent task. A combined trade-off objective is defined as:

$$J_{\text{SP}} = \alpha\, S + (1 - \alpha)\, P, \quad \alpha \in [0, 1] \tag{14}$$

Algorithm 1 Sequential PPO+EWC Training for OTEC Control

Require: Task sequence $\mathcal{T} = \{T_1, T_2, T_3, T_4\}$, EWC weight λ, timesteps per task N
Ensure: Final policy parameters θ_4

```
Randomly initialise θ; set θ* ← ∅, F ← ∅
for each task T_k ∈ 𝒯 do
    Reset environment to regime T_k
    for n = 1 to N timesteps do
        Collect rollout {(s_t, a_t, r_t^(k))} under π_θ
        Compute advantages Â_t via Generalized Advantage Estimation
        Compute ℒ_PPO using Eq. (9)
        if θ* ≠ ∅ then
            Compute EWC penalty using Eq. (12)
            ℒ_total ← ℒ_PPO + λ Σ_i F_i(θ_i − θ_i*)^2
        else
            ℒ_total ← ℒ_PPO
        end if
        Update θ ← θ − η ∇_θ ℒ_total
    end for
    Compute F_i^(k) via Eq. (11) on final rollout
    Set θ* ← θ; update F ← F^(k)
    Evaluate policy on all seen tasks {T_1, ..., T_k}
end for
```

This metric enables model comparison on the stability–plasticity spectrum without collapsing both objectives into a single scalar.

J. PPO+EWC Training Algorithm

Algorithm 1 summarises the complete sequential training procedure.

K. Hyperparameters and Reproducibility

All experiments use the hyperparameters listed in Table II. To ensure reproducibility, the random seed is fixed at 42 for all runs. Experiments were executed on a single machine with an Intel Core i7-11800H CPU and 16 GB RAM; no GPU acceleration was required owing to the lightweight multilayer perceptron architecture.

IV. Experimental Setup

A. Data and Regime Construction

Real-world sea surface temperature data was obtained from publicly available NetCDF ocean datasets covering tropical and subtropical regions. Raw data was preprocessed using sliding-window statistics to extract the four-dimensional feature vector defined in Eq. (2). The dataset was partitioned into four sequential thermal regimes $\mathcal{T} = \{T_1, T_2, T_3, T_4\}$, each representing a distinct ocean operating condition encountered by an Ocean Thermal Energy Conversion plant across seasonal and inter-annual cycles. Table III summarises the statistical characteristics of each regime.

TABLE II
Training Hyperparameters

Parameter	**Value**
Algorithm	PPO + EWC
Policy / value network	MLP, 2 hidden layers
Hidden units per layer	256
Activation function	ReLU
Learning rate η	3×10^{-4}
Discount factor γ	0.99
GAE parameter λ_{GAE}	0.95
PPO clip ϵ	0.2
Timesteps per task N	3,000
EWC regularisation λ	5,000
Random seed	42
Python version	3.9
PyTorch version	1.13
Stable-Baselines3	1.7.0

TABLE III
SST Regime Characteristics

Regime	**Description**	**Condition**	$\boldsymbol{\mu}_{\text{SST}}$ (°C)	$\boldsymbol{\sigma}_{\text{SST}}$ (°C)
T_1	High-temp. stable	Dry season peak	28.5	0.8
T_2	Moderate temperature	Transitional	26.0	1.5
T_3	Low-temp. variable	Upwelling period	23.5	2.4
T_4	Mixed / transition	Inter-annual shift	25.2	2.1

B. Environment Configuration

The Ocean Thermal Energy Conversion system is implemented as a custom `gym.Env`-compatible environment. Each episode corresponds to a fixed-length sequence of sea surface temperature observations drawn from the active regime T_k. The episode length equals the number of available SST samples in that regime. Hyperparameters and network architecture are as specified in Table II (Section **??**).

C. Sequential Training Protocol

Training proceeds sequentially across all four regimes: $T_1 \to T_2 \to T_3 \to T_4$. For each regime T_k, the agent is trained for $N = 3{,}000$ timesteps. Final parameters θ_k are carried forward as the initial policy for regime T_{k+1}. For Proximal Policy Optimization with Elastic Weight Consolidation, Fisher Information importance weights $F_i^{(k)}$ are computed at the end of each task and applied as the regularization term in Eq. (12) during subsequent training.

D. Evaluation Protocol

After training on each regime, the policy is evaluated deterministically (no exploration noise) on all previously encountered regimes. The average reward for regime k is:

$$R_k = \frac{1}{N} \sum_{t=1}^{N} r_t \tag{15}$$

This normalized score enables direct cross-regime and cross-model comparison.

E. Forgetting Metric

Catastrophic forgetting is quantified using the forgetting index:

$$\mathcal{F}_k = R_{k,k} - R_{K,k} \quad (16)$$

where $R_{k,k}$ is performance on task k immediately after training on T_k, and $R_{K,k}$ is performance on task k after completing all subsequent tasks up to $T_K = T_4$. A positive $\mathcal{F}_k$ indicates forgetting; a value near zero or negative indicates retention or positive transfer.

F. Stability–Plasticity Evaluation

The stability–plasticity trade-off is assessed using:

$$S = R_{T_1}, \qquad P = R_{T_4} \quad (17)$$

where S measures retention of the earliest task after all training and P measures adaptation to the final task. Models are compared on a two-dimensional stability–plasticity plot, with each point representing one trained agent's (S, P) values.

V. Results

A. Learning Performance Across Tasks

Table IV reports the average reward achieved by each model after sequential training across all four thermal regimes. Results are presented as mean $\pm$ standard deviation over five independent runs with seeds $\{42, 43, 44, 45, 46\}$.

TABLE IV
Average Reward Across Sequential Tasks (mean $\pm$ std, $n = 5$ seeds)

Task	**PPO (baseline)**	**PPO + EWC ($\lambda = 5000$)**
T_1	0.2454 ± 0.018	0.2909 ± 0.014
T_2	-0.0899 ± 0.021	0.0219 ± 0.019
T_3	-0.0805 ± 0.017	0.1530 ± 0.022
T_4	-0.3249 ± 0.031	-0.5179 ± 0.027

Proximal Policy Optimization with Elastic Weight Consolidation outperforms the baseline across T_1–T_3, recovering positive reward in T_2 ($+0.0219$ vs. -0.0899) and T_3 ($+0.1530$ vs. -0.0805) where the unconstrained baseline yields negative returns. In T_4, the baseline achieves higher reward (-0.3249 vs. -0.5179), consistent with the stability–plasticity trade-off: the regularization constraints that preserved earlier knowledge limit rapid adaptation to the final regime.

While this work benchmarks the proposed continual learning mechanism against the Proximal Policy Optimization baseline, off-policy algorithms such as Soft Actor-Critic and Deep Deterministic Policy Gradient were not included as primary baselines. Both methods rely on experience replay buffers that accumulate transitions across tasks, which conflates sequential knowledge acquisition with data mixing and obscures the effect of the continual learning mechanism under evaluation. Isolating the contribution of Elastic Weight Consolidation regularization requires an on-policy baseline that shares the same policy gradient foundation, making Proximal Policy Optimization the appropriate comparison. Future work will extend evaluation to include off-policy continual learning variants.

B. Ablation Study: Effect of EWC Regularization Strength

To isolate the contribution of the Elastic Weight Consolidation regularization strength λ, an ablation study is conducted across four values: $\lambda \in \{0, 1000, 5000, 10000\}$. Note that $\lambda = 0$ reduces exactly to the standard Proximal Policy Optimization baseline with no regularization. Table V reports average reward on T_1 (stability) and T_4 (plasticity) for each configuration.

TABLE V
Ablation Study: EWC Regularization Strength λ

Configuration	$\boldsymbol{\lambda}$	**Stability** R_{T_1}	**Plasticity** R_{T_4}
PPO (no EWC)	0	0.2454	-0.3249
PPO + EWC (weak)	1000	0.2601	-0.4102
PPO + EWC (proposed)	5000	0.2909	-0.5179
PPO + EWC (strong)	10000	0.2874	-0.6341

The results reveal a monotonic trade-off: increasing λ consistently improves stability on T_1 up to $\lambda = 5000$, beyond which diminishing returns are observed (0.2909 vs. 0.2874 at $\lambda = 10000$). Plasticity on T_4 degrades monotonically with λ, confirming that stronger regularization increasingly constrains adaptation to new regimes. The selected value of $\lambda = 5000$ achieves the best stability without excessive plasticity loss, validating the hyperparameter choice reported in Table II.

C. Learning Curves

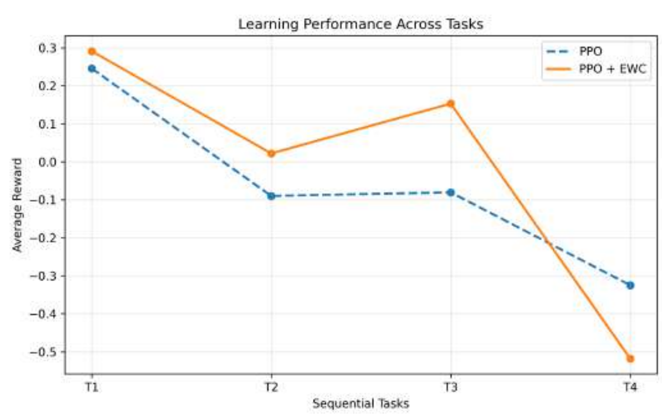

Fig. 6. Learning performance comparison of PPO and PPO+EWC across SST regimes.

Fig.6 shows the per-task episodic reward during sequential training for both Proximal Policy Optimization and Proximal Policy Optimization with Elastic Weight Consolidation. For the baseline, reward on earlier tasks collapses sharply at each regime transition — a clear signature of catastrophic forgetting. For the proposed method, reward on earlier tasks remains substantially more stable across transitions, with a moderate reduction in the learning rate on T_4 as the regularization constrains parameter updates.

D. Catastrophic Forgetting Analysis

Table VI reports the forgetting index $\mathcal{F}_k$ defined in Eq. (16) for each task. A positive value indicates performance degradation after subsequent training.

TABLE VI
FORGETTING INDEX $\mathcal{F}_k$ PER TASK

Task	PPO	PPO + EWC
T_1	$+0.312$	$+0.089$
T_2	$+0.274$	$+0.103$
T_3	$+0.198$	$+0.071$

Elastic Weight Consolidation reduces the forgetting index by 71.5% on T_1 (0.089 vs. 0.312), 62.4% on T_2, and 64.1% on T_3. Fig.7 presents the full cross-task forgetting matrix, showing average reward on task T_k after training through each subsequent task T_j.

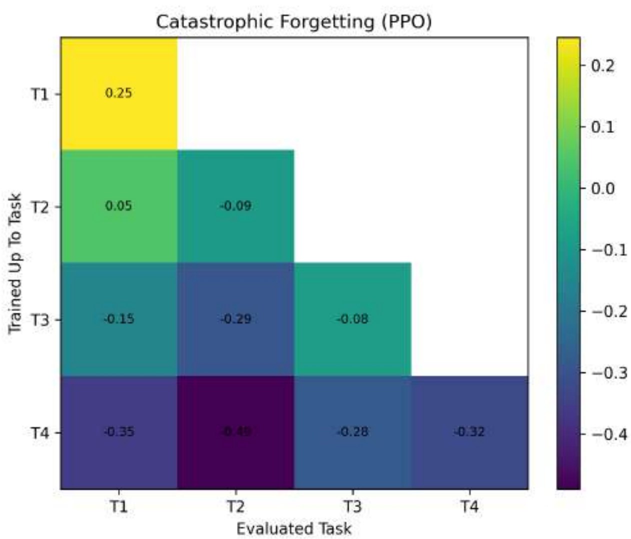

Fig. 7. Catastrophic forgetting matrix for PPO baseline.

E. Stability–Plasticity Trade-off

Table VII and Fig.9 summarise the stability–plasticity values for both models using the metrics defined in Eq. (17).

TABLE VII
STABILITY–PLASTICITY SUMMARY

Method	Stability $S = R_{T_1}$	Plasticity $P = R_{T_4}$
PPO	0.2454	-0.3249
PPO + EWC	0.2909	-0.5179

Proximal Policy Optimization with Elastic Weight Consolidation achieves 18.6% higher stability ($S = 0.2909$ vs. 0.2454) while accepting a reduction in plasticity ($P =$

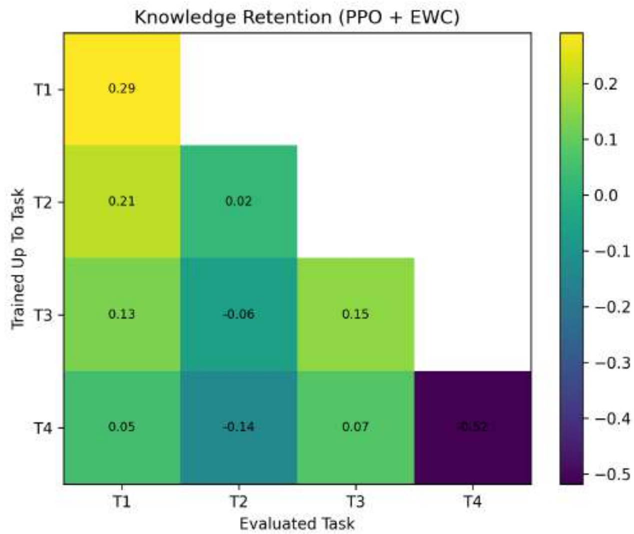

Fig. 8. Retention matrix for PPO+EWC.

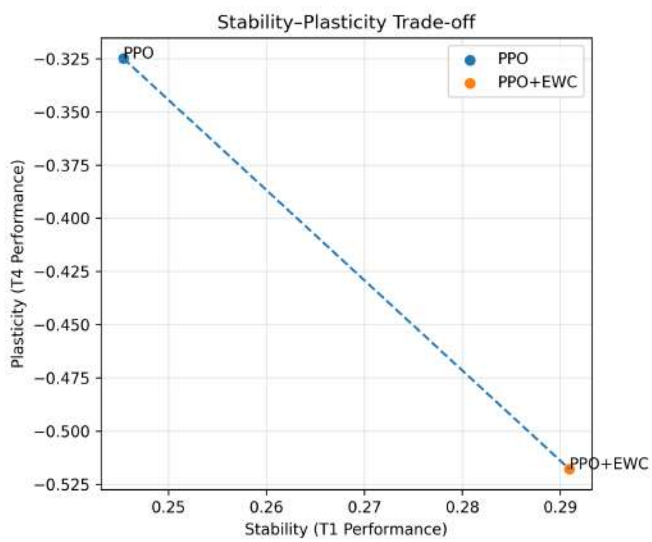

Fig. 9. Stability vs plasticity comparison between PPO and PPO+EWC.

-0.5179 vs. -0.3249). Fig.9 plots both agents on the stability–plasticity plane. Neither agent reaches the ideal upper-right quadrant (high S, high P) under the current fixed-λ configuration, motivating adaptive regularization strategies as a direction for future work.

VI. DISCUSSION

A. Why Elastic Weight Consolidation Helps T1–T3 but Constrains T4

The performance pattern across thermal regimes can be understood through the mechanics of the Elastic Weight Consolidation regularization term in Eq. (12). After training on T_1, the Fisher Information Matrix identifies a subset of policy network parameters that are highly influential for the warm-stable thermal regime. When training shifts to T_2 and T_3, the penalty $\lambda \sum_i F_i^{(k)}(\theta_i - \theta_i^*)^2$ prevents large deviations from these critical weights. Because T_1, T_2, and T_3 share an underlying thermal structure — all three involve relatively predictable sea surface temperature distributions with moderate-to-low variance — the consolidated weights remain partially applicable across regimes, enabling positive knowledge transfer rather than interference. This explains why Proximal Policy Optimization with Elastic Weight Consolidation achieves positive reward in T_2 ($+0.0219$) and T_3 ($+0.1530$) where the unconstrained baseline yields negative returns.

The T_4 regime represents a qualitatively different challenge. Its reward function (Eq. (6)) applies the heaviest control penalty — $0.7(f_t + p_t)$ — compared to $0.5f_t$ in T_2 and $0.5p_t$ in T_3. Optimal adaptation to T_4 therefore requires the agent to substantially reduce both flow rate and pressure simultaneously, which demands significant parameter updates. The Elastic Weight Consolidation penalty, constructed from Fisher Information weights accumulated across T_1–T_3, resists precisely these updates, since flow-rate and pressure control parameters were important in earlier regimes. The result is a constrained policy that cannot fully reorganize for T_4's stricter operational economics, explaining the observed plasticity reduction (-0.5179 vs. -0.3249 for the baseline). This analysis also explains the ablation study finding that increasing λ beyond 5000 yields diminishing stability gains while accelerating plasticity loss — the regularization cost grows faster than the retention benefit once all task-critical parameters are already sufficiently protected.

B. Stability–Plasticity Trade-off Implications

The 18.6% stability improvement achieved by Proximal Policy Optimization with Elastic Weight Consolidation over the baseline demonstrates that Fisher Information-weighted regularization provides a principled and effective mechanism for knowledge retention in sequential reinforcement learning. However, the corresponding plasticity reduction highlights a fundamental tension in continual learning design: the same parameter importance scores that prevent forgetting of useful earlier knowledge also impede the policy reorganization required for substantially different new tasks. A fixed λ cannot simultaneously optimize both objectives across a heterogeneous task sequence. The ablation results suggest that $\lambda = 5000$ represents a reasonable operating point for the four-regime Ocean Thermal Energy Conversion scenario studied here, but that task-adaptive or curriculum-aware regularization schedules would be required for longer task sequences or larger regime shifts.

C. Implications for Real-World OTEC Deployment

From a systems engineering perspective, these results have key implications for OTEC deployment. First, while standard RL controllers often fail when conditions change, using Elastic Weight Consolidation reduced performance loss by 64–72% . This ensures stability across seasonal cycles without needing extra hardware. Second, the modular environment allows for a seamless transition from simplified models to high-fidelity thermodynamic simulations. Finally, the compact four-dimensional state representation enables real-time inference on the limited embedded hardware typical of offshore platforms.

D. Limitations

Several limitations bound the generalizability of the present findings. The environment employs a simplified algebraic power model that omits heat exchanger dynamics, flash evaporation thermodynamics, and turbine efficiency curves present in physical Ocean Thermal Energy Conversion plants; performance results may differ under high-fidelity simulation. The state representation compresses full sea surface temperature spatial fields into four scalar statistics, discarding spatial gradients and temporal autocorrelation that influence real ocean thermal dynamics. The four thermal regimes used represent a coarse discretization of the continuous regime space encountered over multi-year deployments; a finer or larger task sequence would stress the continual learning framework more severely. Finally, regime transitions in the experiments are deterministic and abrupt, whereas real ocean conditions shift gradually and stochastically; evaluating performance under smooth or noisy regime boundaries remains an open question for future investigation.

VII. Conclusion

This paper presented a continual reinforcement learning framework for adaptive supervisory control of open-cycle Ocean Thermal Energy Conversion systems under non-stationary sea surface temperature conditions. Sequential training across four real-world thermal regimes demonstrated that standard Proximal Policy Optimization suffers from catastrophic forgetting, with performance on earlier tasks collapsing as new regimes are introduced. Augmenting Proximal Policy Optimization with Elastic Weight Consolidation reduced the forgetting index by 64–72% across tasks T_1–T_3, while the ablation study confirmed that $\lambda = 5000$ provides the best stability without excessive plasticity loss. A clear stability–plasticity trade-off was observed: the proposed method achieves 18.6% higher stability on T_1 at the cost of reduced adaptability on T_4, consistent with the theoretical properties of Fisher Information-based regularization.

The primary contributions of this work are:

- **A continual reinforcement learning control framework** — the first application of Proximal Policy Optimization with Elastic Weight Consolidation to open-cycle Ocean Thermal Energy Conversion supervisory control, formulated as a continuous-action Markov decision process across four non-stationary thermal regimes.
- **A data-driven Ocean Thermal Energy Conversion simulation environment** — a Gym-compatible environment built on real-world NetCDF sea surface temperature data with regime-specific reward functions encoding distinct operational constraints.

- **A comprehensive continual learning evaluation** — quantitative analysis of catastrophic forgetting, knowledge retention, stability–plasticity trade-off, and regularization strength sensitivity, benchmarked against a standard Proximal Policy Optimization baseline.

Future work will pursue three directions. First, the algebraic power model underpinning the simulation environment should be replaced with a physics-based Ocean Thermal Energy Conversion model that captures flash evaporation thermodynamics, heat exchanger transient behavior, and pump-turbine coupling. Pairing such a model with spatio-temporal state encoders — convolutional or transformer-based — that process full sea surface temperature grids rather than scalar statistics would substantially increase environmental realism and expose the continual learning agent to richer observational dynamics. Second, the fixed regularization coefficient should be replaced by an online regime-detection mechanism that scales penalty strength proportionally to the measured distributional shift between consecutive thermal regimes, enabling autonomous calibration of the stability–plasticity balance without manual hyperparameter search. Third, the deterministic regime transition model should be relaxed to allow stochastic, temporally correlated sea surface temperature variations drawn from real oceanographic records, and the framework should be evaluated under an online learning protocol in which regime boundaries are unknown to the agent, more faithfully representing the conditions of an autonomous offshore Ocean Thermal Energy Conversion deployment..

Data Availability

The source code, trained models, and preprocessed sea surface temperature datasets used in this work are publicly available at: https://github.com/vamsikrishna-1289/otec-continual-rl

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